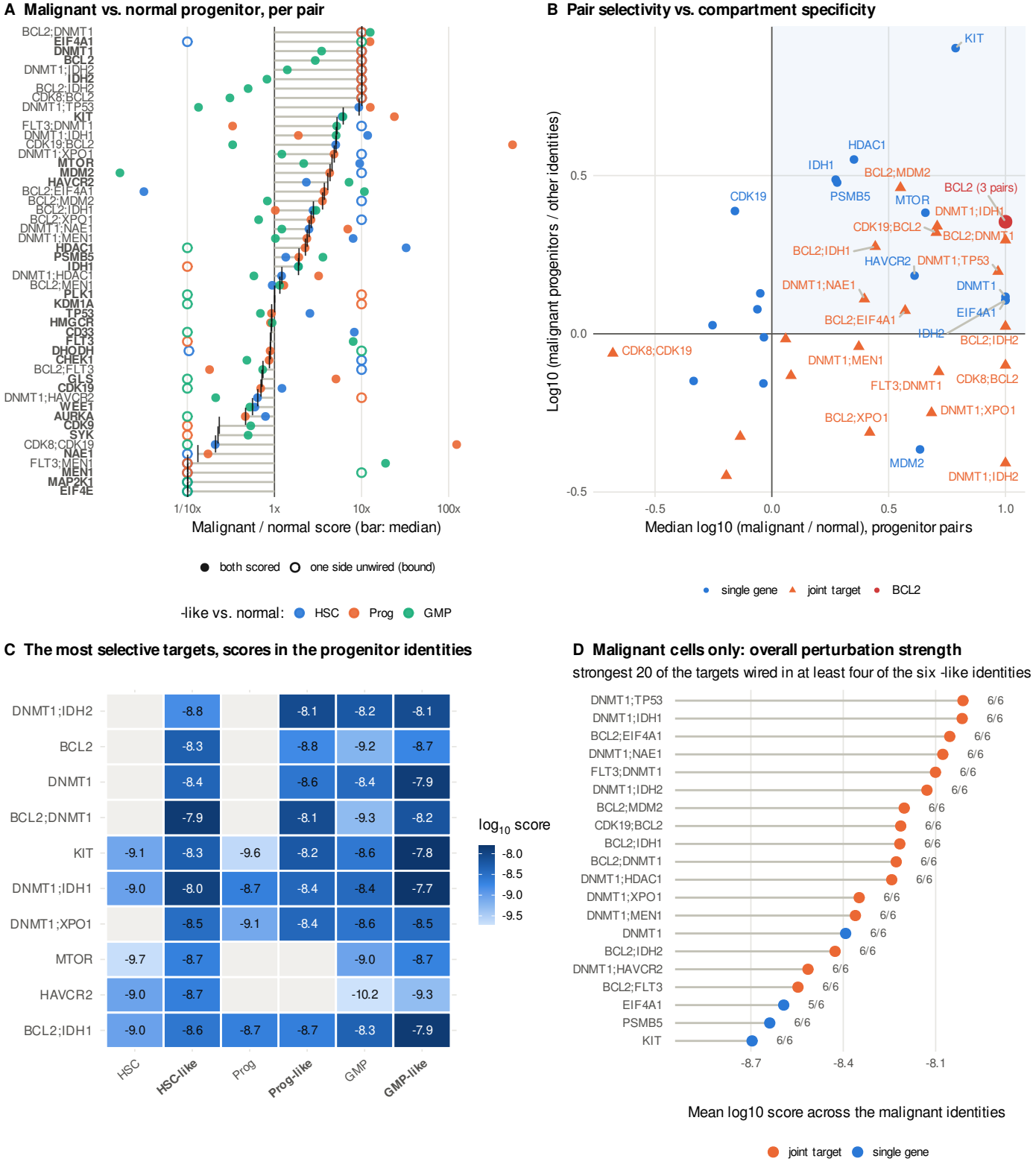
A Malignant vs. normal progenitor, per pair
BCL2;DNMT1
EIF4A1
DNMT1
BCL2
DNMT1;IDH2
IDH2
BCL2;IDH2
CDK8;BCL2
DNMT1;TP53
KIT
FLT3;DNMT1
DNMT1;IDH1
CDK19;BCL2
DNMT1;XPO1
MTOR
MDM2
HAVCR2
BCL2;EIF4A1
BCL2;MDM2
BCL2;IDH1
BCL2;XPO1
DNMT1;NAE1
DNMT1;MEN1
HDAC1
PSMB5
IDH1
DNMT1;HDAC1
BCL2;MEN1
PLK1
KDM1A
TP53
HMGCR
CD33
FLT3
DHODH
CHEK1
BCL2;FLT3
GLS
CDK19
DNMT1;HAVCR2
WEE1
AURKA
CDK9
SYK
CDK8;CDK19
NAE1
FLT3;MEN1
MEN1
MAP2K1
EIF4E
1/10x
1x
10x
100x
Malignant / normal score (bar: median)
both scored
one side unwired (bound)
-like vs. normal:
HSC
Prog
GMP
B Pair selectivity vs. compartment specificity
Log10 (malignant progenitors / other identities)
0.5
0.0
-0.5
-0.5
0.0
0.5
1.0
Median log10 (malignant / normal), progenitor pairs
KIT
HDAC1
IDH1
BCL2;MDM2
PSMB5
CDK19
MTOR
BCL2 (3 pairs)
DNMT1;IDH1
CDK19;BCL2
BCL2;DNMT1
BCL2;IDH1
HAVCR2
DNMT1;TP53
DNMT1;NAE1
DNMT1
BCL2;EIF4A1
EIF4A1
IDH2
BCL2;IDH2
CDK8;CDK19
DNMT1;MEN1
FLT3;DNMT1
CDK8;BCL2
BCL2;XPO1
DNMT1;XPO1
MDM2
DNMT1;IDH2
single gene
joint target
BCL2
C The most selective targets, scores in the progenitor identities
DNMT1;IDH2
-8.8
-8.1
-8.2
-8.1
BCL2
-8.3
-8.8
-9.2
-8.7
DNMT1
-8.4
-8.6
-8.4
-7.9
BCL2;DNMT1
-7.9
-8.1
-9.3
-8.2
KIT
-9.1
-8.3
-9.6
-8.2
-8.6
-7.8
DNMT1;IDH1
-9.0
-8.0
-8.7
-8.4
-8.4
-7.7
DNMT1;XPO1
-8.5
-9.1
-8.4
-8.6
-8.5
MTOR
-9.7
-8.7
-9.0
-8.7
HAVCR2
-9.0
-8.7
-10.2
-9.3
BCL2;IDH1
-9.0
-8.6
-8.7
-8.7
-8.3
-7.9
HSC
HSC-like
Prog
Prog-like
GMP
GMP-like
log10 score
-8.0
-8.5
-9.0
-9.5
D Malignant cells only: overall perturbation strength
strongest 20 of the targets wired in at least four of the six -like identities
DNMT1;TP53
DNMT1;IDH1
BCL2;EIF4A1
DNMT1;NAE1
FLT3;DNMT1
DNMT1;IDH2
BCL2;MDM2
CDK19;BCL2
BCL2;IDH1
BCL2;DNMT1
DNMT1;HDAC1
DNMT1;XPO1
DNMT1;MEN1
DNMT1
BCL2;IDH2
DNMT1;HAVCR2
BCL2;FLT3
EIF4A1
PSMB5
KIT
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
6/6
5/6
6/6
6/6
-8.7
-8.4
-8.1
Mean log10 score across the malignant identities
joint target
single gene